

Energy management strategy of fuel cell electric vehicles using learning-based EMPC

Jinghan Cui, Shengyan Hou, Zheng Zhu, Yuqi Liu, Jinwu Gao

Abstract—A reliable energy management strategy (EMS) incorporating vehicle connectivity technology is of great importance for the performance enhancement of fuel cell electric vehicles (FCEVs). Cloud computing and learning-based methods create new opportunities and challenges for EMS by offering powerful computation and storage capabilities. To fully utilize connected driving information and maintain closed-loop economic performance, while reducing the computational burden, this paper proposes a novel cloud-assisted economic model predictive control with learning-based terminal cost (EMPC-LTC), which is tailored to the EMS of FCEVs. The economic model predictive control (EMPC) optimizations are performed both on the cloud and on the local processor. On the cloud side, a long-horizon EMPC with future driving condition is constructed. Then, the cloud can support a learning process of a parametric terminal cost acting as a good approximation of the cost-to-go performance of the long-horizon EMPC. On the local agent side, the onboard controller runs a short-horizon EMPC with the learning-based terminal cost obtained from the cloud, which can reduce online computing. The effectiveness of this EMPC-LTC is verified through the probabilistic validation technique, considering the errors caused by transmission and learning. An application of this proposed controller on the EMSs of FCEVs demonstrates its effectiveness.

Index Terms—Economic model predictive control, fuel cell electric vehicle, neural networks, cloud computing, dynamic economy.

I. INTRODUCTION

At present, the aggressive exploitation of fossil fuels combined with rising energy demand has led to significant greenhouse gas emissions and environmental degradation, prompting extensive research into energy-saving and emission-reduction technologies [1]. For the road transportation industry, fuel cell electric vehicles (FCEVs), as one of the most promising types of new energy vehicles, are emerging as a research hotspot, driven by their zero-emission advantages and their effective integration with electric energy storage systems [2]. In FCEVs, two or more energy sources are assigned either independently or in combination to deliver

driving power. Hence, it allows for flexible energy management and optimization to reduce fuel consumption [3]. The complexities involved in multiple energy pathways pose significant challenges, making the development of effective energy management strategies (EMSs) a critical area for both researchers and industry practitioners. Besides, due to the development of connected vehicle technologies, such as vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communication, provide new opportunities for EMSs since they can offer information about driving conditions over a period of time [4], [5]. As a result, designing an optimal and real-time EMS for FCEVs that efficiently utilizes connected traffic information has become a critical need.

The objectives for EMS are to decrease fuel consumption and emissions while extending the lifespan of energy storage systems [6]. In existing studies, popular EMSs are generally categorized into two main approaches: rule-based approaches and optimization-based approaches [7]. Although rule-based strategies are simple to design and implement, their optimality and adaptability can not be achieved since these methods mainly rely on engineering experience. Optimal control serves as an appropriate optimization-based framework to achieve the EMS objectives. Then, the energy management task can be framed as a constrained nonlinear dynamic optimization since the powertrain in FCEV exhibits nonlinear behavior and the states and control inputs are subject to multiple equality and inequality limits.

In the literatures, various advanced EMSs based on optimal control framework have been investigated for FCEVs. Model predictive control (MPC), as a preferable method to approximate the infinite-horizon optimal control, possesses the capacity to incorporate the input and state constraints into an optimization problem and achieve optimality explicitly using a receding horizon mechanism [8]. The MPC-based EMS is often embedded in a hierarchical structure, i.e., hierarchical MPC (HMPC) [9], [10]. The upper layer of HMPC calculates the reference trajectory of the battery state of charge (SOC), which will be sent to the lower layer, while the lower layer achieves the tracking of the trajectory and the satisfaction of the constraints. However, this hierarchical structure may bring about the loss of dynamic economic performance in the tracking process. The lower layer is unable to access complete future driving conditions, which prevents the achievement of optimality. Moreover, the interaction of information between the two layers introduces additional uncertainty and disturbances, leading to the deterioration of economic performance.

Economic model predictive control (EMPC) has been investigated as a potential technique since it can maximize energy efficiency by incorporating the reference trajectory

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optimization and tracking MPC into a single optimal control framework [11]. The closed-loop economic performance of EMPC has been proven to be better than traditional H MPC in many application areas [12]–[14]. For EMS of FCEVs, EMPC is a promising approach since it is now possible to preview vehicle speed over a longer prediction horizon which should be made full use of to guarantee closed-loop economic performance. However, a sufficiently large prediction horizon is not suitable for real-time optimization as it results in a significant on-line computational complexity.

In recent years, the rapid advancement of artificial intelligence (AI) has led to the growing exploration and application of machine learning (ML) techniques in the field of energy management. Both reinforcement learning (RL) and deep learning (DL) can be introduced into the energy management for FCEVs to improve economic performance and reduce operating costs. Among them, a speed prediction constructed based on DL is always incorporated into the optimization of EMS, enabling it to approach global optimization [15]. In addition, RL-based EMSs can leverage value functions to assess the long-term expected reward of taking actions from a given state, thereby facilitating effective long-horizon decision-making. RL strategies are well-suited for managing complex nonlinear dynamics and offer the potential to achieve globally optimal performance [16]. However, the practical deployment of RL is often constrained by low sample efficiency and limited generalization capability. Therefore, finding an energy management strategy that achieves near-global optimization with real-time performance still remains a significant challenge.

To make full use of the future driving conditions, preserving desirable closed-loop economic performance while reducing the computational burden for EMS by designing an EMPC strategy assisted with learning methods has motivated this work. Previous studies have employed machine learning to develop predictive models [12] or to approximate explicit MPC solutions [17]. However, both approaches offer limited support for enabling EMPC to achieve global optimization. To address this limitation, an infinite-horizon EMPC framework incorporating terminal cost learning is introduced into the EMS design to enhance its optimality and real-time performance. The terminal cost always represents the upper bound of the closed-loop performance over an infinite horizon under a locally stable controller [18]. Securing an effective balance between closed-loop optimality and real-time computational burden hinges on judiciously selecting the prediction horizon and designing a learning-based terminal cost. One method to obtain a non-conservative terminal cost is to resort to the data-driven approaches [19].

In this work, a novel cloud-assisted economic model predictive control with learning-based terminal cost (EMPC-LTC) is tailored to the EMS of FCEVs. The key contributions of this work are summarized as:

- 1) A novel cloud-assisted control architecture is developed, where a long-horizon EMPC is executed on the cloud using previewed driving information, while a short-horizon EMPC runs locally on the onboard controller. This coordination enables a balance between closed-loop economy and real-time performance.

- 2) A parameterized terminal cost function is learned offline on the cloud to approximate the cost-to-go of the long-horizon EMPC. This learned terminal cost is transmitted to the local controller and incorporated into the short-horizon EMPC, substantially reducing online computational burden while retaining near-optimality.
- 3) A probabilistic validation procedure is introduced to assess the robustness of the proposed EMPC-LTC framework. The effectiveness of the method is demonstrated through its application to the EMS of a FCEV, confirming its capability to ensure both economic performance and real-time implementability.

The rest of this paper is organized as follows. The model description of the FCEV system and problem formulation are presented in Section II. In Section III, the proposed EMPC strategy for EMS is explained clearly. Section IV gives a detailed evaluation of the proposed EMPC via experiments. Finally, Section V concludes this paper.

II. PRELIMINARIES

A. Model description of the FCEV system

The architecture of the FCEV powertrain is depicted in Figure 1. The fuel cell system (FCS) and battery pack are two main components, and FCS supplies the powertrain's principal energy linked to the DC bus via a unidirectional DC/DC converter, whereas the battery pack acts as a secondary power source connected to the DC bus to meet peak power demands and capturing regenerative braking energy. The power provided by the FCS (P_F) and the battery pack (P_B) is blended to meet the power demand (P_D) of the vehicle for driving, shown as

$$P_D(t) = P_F(t) + P_B(t) - P_A(t), \quad (1)$$

where $P_A(t)$ is the power for auxiliary systems, and t denotes time. The power demand is related to the driving conditions, which can be formulated as

$$P_D(t) = \frac{F(t)v(t)}{\eta}, \quad (2)$$

where $F(t)$ represents the driving force, $v(t)$ is the vehicle speed, and η is the mechanical transmission efficiency. The vehicle's longitudinal dynamics model, which governs the computation of traction force, serves as the foundation for analyzing vehicle performance. During the operation of a FCEV, the motor's output torque is transmitted to the wheels via the drivetrain, resulting in a driving force $F(t)$ that counterbalances the resistance encountered by the vehicle. The driving force $F(t)$ is calculated by

$$F(t) = mgf \cos \alpha + mg \sin \alpha + \frac{1}{2} \rho A c_d v^2 + m\dot{v}, \quad (3)$$

where m represents the mass of the FCEV, g is the gravitational acceleration, f is the rolling resistance coefficient, c_d is the aerodynamic drag coefficient, A is the windward area, ρ is the air density, and α is the road slope.

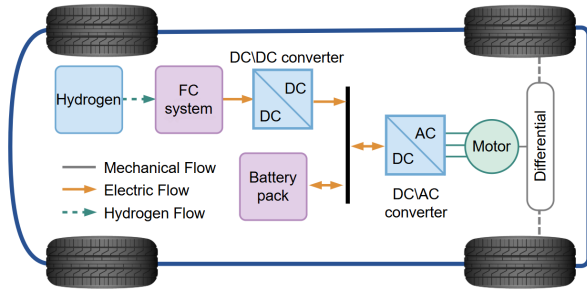


Fig. 1. The FCEV configuration.

For the battery pack, an internal resistance model is adopted to characterise its behaviour, neglecting the influence of temperature effects. The Rint model is employed to represent the battery, and the output power P_B can be written as,

$$P_B = U_B I_B - R I_B^2 \quad (4)$$

where I_B , U_B , and R are the battery current, open-circuit voltage, and internal resistance respectively. The battery current derived from the equivalent circuit model depends on the output power, internal resistance, and open-circuit voltage, and is given by,

$$I_B = \frac{U_B - \sqrt{U_B^2 - 4RP_B}}{2R}. \quad (5)$$

Then, the battery's SOC, as the main state in the EMS of a FCEV, can be modelled through an equivalent-circuit representation

$$\dot{SOC}(t) = f_{SOC}(t) = \frac{U_B - \sqrt{U_B^2 - 4RP_B}}{2RQ_B}, \quad (6)$$

where Q_B is the battery capacity. The battery's numerical modelling characteristics are illustrated in Figure 2 and Figure 3. To ensure the continuity of the model, three continuous functions are constructed to fit the data relationships, expressed as,

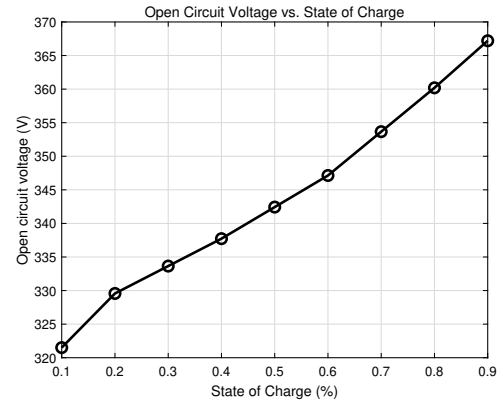
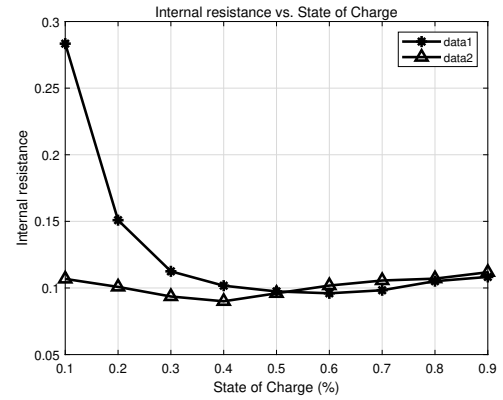
$$U_B = N_c \sum_{i=0}^6 \alpha_i^{U_B} \times (SOC)^i \quad (7a)$$

$$R_{dis} = N_c \sum_{i=0}^8 \alpha_i^{dis} \times (SOC)^i \quad (7b)$$

$$R_{cha} = N_c \sum_{i=1}^3 a_i \sin(b_i \times SOC + c_i) \quad (7c)$$

where N_c is the number of batteries in series. The positive battery power, i.e., $P_B > 0$, denotes that the battery is discharging with $R = R_{dis}$, while negative battery power, i.e., $P_B < 0$, denotes that the battery is charging with $R = R_{cha}$.

As for FCS, a static model rather than any dynamic behavior is considered in this work since it mainly focuses on fuel cell energy optimization. An efficiency model and a hydrogen consumption model for the FCS are involved in the optimization problem. Moreover, hydrogen usage for the FCS is inferred

Fig. 2. The open circuit voltage U_B .Fig. 3. The internal resistance R .

from its output power together with the associated efficiency, expressed as

$$\dot{m}_F(t) = \frac{1}{E_{H_2}} \int \frac{P_F(\tau)}{\eta_F(P_F(\tau))} d\tau, \quad (8)$$

where $\dot{m}_F$ stands for the hydrogen consumption of the FCS, E_{H_2} is the model parameter representing the lower heating value of hydrogen, and η_F is the efficiency as a function of the fuel cell power P_F . A convex function for hydrogen consumption model can be obtained using fitting methods with the measured data from experiments, expressed as [9],

$$\dot{m}_F(t) = \sum_{i=0}^2 \alpha_i^m \times (P_F)^i. \quad (9)$$

Furthermore, the following physical constraints must be enforced,

$$SOC_{min} \leq SOC(t) \leq SOC_{max}, \quad (10a)$$

$$0 \leq P_F(t) \leq P_{F,max}, \quad (10b)$$

$$P_{B,min} \leq P_B(t) \leq P_{B,max}, \quad (10c)$$

where min and max denote the minimum and maximum bounds for system variables.

Based on the previous equations and descriptions, define the state variable as $x = SOC$, the input variable as $u = P_F$, the known demand as $p = P_D$ which is calculated by the measured

TABLE I
THE PARAMETERS OF THE FCEV.

Parameter	Symbol	Value	Unit
Auxiliary power	P_A	1	kW
Vehicle basis mass	m	1700	kg
Rolling resistance coefficient	f	0.0074	-
Aerodynamic drag coefficient	c_d	0.371	-
Windward area	A	2.65	m^2
Air density	ρ	1.1839	kg/m^3
Battery capacity	Q_B	12	Ah
Heating value of hydrogen	E_{H_2}	120	kJ/kg
Number of batteries in series	N_c	96	-
Maximum SOC	SOC_{max}	0.8	-
Minimum SOC	SOC_{min}	0.3	-
Fuel cell maximum power	$P_{F,max}$	20	kW
Battery maximum power	$P_{B,max}$	70	kW
Battery minimum power	$P_{B,min}$	-70	kW
Fitting parameters for U_B	$\alpha_i^{U_B}, i = 0, \dots, 6$	3.12, 3.54, -16.83, 46.04, -67.98, 51.95, -15.94	-
Fitting parameters for R_{dis}	$\alpha_i^{dis}, i = 0, \dots, 8$	14.79, -147.5, 801.9, -2663, 5855, -8584, 8022, -4276, 979.7	-
Fitting parameters for R_{cha}	$[a_i, b_i, c_i], i = 1, 2, 3$	[0.06, 14.4, -2.3], [2.1, 2.9, 3.5], [4, 1.6, 1]	-
Fitting parameters for $\dot{m}_F$	$\alpha_i^m, i = 0, \dots, 2$	0.02, 1.01e-5, 2e-10	-

vehicle speed, the output variables as $y = [\dot{m}_F \ P_B]^T$. Then, the control-oriented model may be expressed succinctly as follows

$$\dot{x}(t) = f(x(t), u(t), p(t)), \quad (11a)$$

$$y(t) = g(x(t), u(t), p(t)). \quad (11b)$$

The related parameters can refer to the Table I, provided by Dongfeng Motor Corporation.

B. Problem formulation

To make full use of future driving conditions, preserving desirable closed-loop economic performance while reducing the computational burden, an effective EMPC-based EMS should be tailored. Multiple energy sources and distinct components within the powertrain in FCEV can work in various operating modes to maximize vehicle performance. This multi-source and multi-modes configuration also raise the complexity of the prediction model for powertrain system dynamics. Specifically, the upper and lower bounds of the input variable P_F varies with the demand power P_D due to the constraint of P_B . Besides, the dynamic of the battery's SOC depends on the internal resistance R , which also changes with the P_B . Therefore, both scenarios increase the complexity of the prediction model, leading to an increase in the computational burden associated with solving the optimization problem.

The conventional HMPc is a candidate for lower the computational burden. However, this hierarchical structure may bring about the loss of dynamic economic performance in the tracking process since the lower layer is unable to access complete future driving conditions, which prevents the achievement of global optimality. In the next section, an EMPC strategy assisted with learning methods is elaborated, which secures an effective balance between closed-loop optimality and real-time computational burden via judiciously selecting the prediction horizon and designing a learning-based terminal cost.

III. THE PROPOSED LEARNING-BASED EMPC FOR EMS

In this section, we introduce the long-horizon EMPC on the cloud side and the short-horizon EMPC on the local side.

A. The long-horizon EMPC

When the road condition is assigned, it is assumed that the corresponding preferable driving speed is determined, along with the power demand P_D . The EMPC-based EMS is then designed to minimize hydrogen consumption over a finite prediction horizon while maintaining the battery's state of charge within a charge-sustaining range. Accordingly, the following economic performance indices are defined:

$$l_{e1}(x(t), u(t)) = \dot{m}_F(t), \quad (12a)$$

$$l_{e2}(x(t), u(t)) = (SOC(t) - SOC_r)^2, \quad (12b)$$

where SOC_r is the reference value for the battery. Then, the economic cost function of the EMS is derived as

$$l_e(x(t), u(t)) = \gamma_1 l_{e1}(x(t), u(t)) + \gamma_2 l_{e2}(x(t), u(t)), \quad (13)$$

where γ_1 and γ_2 are the weighted coefficients of l_{e1} and l_{e2} respectively.

Consequently, the long-horizon EMPC optimization problem $\mathbb{P}_N$ executed on the cloud side at time instant t_k is formulated as

$$\min_{u(t) \in S(\Delta)} \sum_{i=0}^{N-1} \int_{t_{k+i}}^{t_{k+i+1}} l_e(\tilde{x}(t), u(t)) dt + V_f(\tilde{x}(t_{k+N})) \quad (14a)$$

$$\text{s.t. } \tilde{x}(t_k) = SOC(t_k), \quad (14b)$$

$$\dot{\tilde{x}}(t) = f(\tilde{x}(t), u(t), p(t)), \quad (14c)$$

$$\dot{\tilde{y}}(t) = g(\tilde{x}(t), u(t), p(t)), \quad (14d)$$

$$SOC_{min} \leq x(t) \leq SOC_{max}, \quad (14e)$$

$$P_{F,min} \leq u(t) \leq P_{F,max}, \quad (14f)$$

$$P_{B,min} \leq y_2(t) \leq P_{B,max}, \quad (14g)$$

$$t \in [t_k, t_{k+N}] \quad (14h)$$

where $S(\Delta)$ designates the class of piecewise-continuous functions defined with sampling period Δ . N is the prediction horizon. $\tilde{x}(t)$ is the predicted SOC trajectory of the FCEV. $SOC(t_k)$ is the initial state at the time t_k . The reference value SOC_r for the battery is always set to be the same as $SOC(t_0)$. A suitable terminal cost is always incorporated into the objective function to replace the infinite-horizon cost function for stability and practicality. Generally, a quadratic terminal cost is employed which means $V_f(\tilde{x}(t_{k+N})) = \gamma_3(x(t_{k+N}) - SOC_r)^2$. All the weighted coefficients γ_1 , γ_2 and γ_3 are positive.

The optimal solution of the optimisation formulation $\mathbb{P}_N$ at time instant t_k is defined as $U_{t_k}^* = \{u_{t_k|k}^*, u_{t_{k+1}|k}^*, \dots, u_{t_{k+N-1}|k}^*\}$, the corresponding optimal state trajectory is defined as $X_{t_k}^* = \{x_{t_k}, x_{t_{k+1}|k}^*, \dots, x_{t_{k+N}|k}^*\}$. According to the receding horizon mechanism, only the current input $u_{t_k}^*$ is employed to generate the optimal manipulated input, i.e., $u_N^*(t|t_k) = u_{t_k}^*, \forall t \in [t_k, t_{k+1}]$.

To ensure system stability and enhance closed-loop economic performance by fully leveraging future driving information, a long prediction horizon is typically required in EMPC design. However, this comes at the cost of considerable computational demand in real-time implementations. To mitigate this issue, an effective strategy is to incorporate a well-designed terminal cost, enabling the use of a shorter prediction horizon without substantial loss in control performance. Leveraging Bellman's principle of optimality, we reformulate the EMPC problem to operate with a short horizon, aiming to approximate the behaviour of the long-horizon EMPC problem $\mathbb{P}_N$ defined in (14). Then, the EMPC problem $\mathbb{P}_{N_s}$ with the shorter predictive horizon N_s at time instant t_k is defined as

$$\min_{u(t) \in S(\Delta)} \sum_{i=0}^{N_s-1} \int_{t_{k+i}}^{t_{k+i+1}} l_e(\tilde{x}(t), u(t)) dt + \quad (15a)$$

$$\mathcal{V}_f(\tilde{x}(t_{k+N_s|k}), p(t_{k+N_s:k+N-1}))$$

$$\text{s.t. } \tilde{x}(t_k) = SOC(t_k), \quad (15b)$$

$$(14c), (14d), \quad (15c)$$

$$u(t) \in \mathcal{U}(p(t)), x(t) \in \mathcal{X}, \quad (15d)$$

$$t \in [t_k, t_{k+N_s}], \quad (15e)$$

where $\mathcal{U}(p(t))$ and $\mathcal{X}$ stand for the admissible input set and state set with respect to (14e-14g). $p(t_{k+N_s:k+N-1|k})$ is the future power demand from time instant t_{k+N_s} to t_{k+N-1} . $\mathcal{V}_f(\cdot)$ is the cost-to-go defined as

$$\mathcal{V}_f(x_{t_{k+N_s|k}}^*, p(t_{k+N_s:k+N-1})) = \min_{u(t) \in S(\Delta)} \sum_{i=N_s}^{N-1} \int_{t_{k+i}}^{t_{k+i+1}} l_e(\tilde{x}(t), u(t)) dt + V_f(\tilde{x}(t_{k+N})) \quad (16a)$$

$$\text{s.t. } \tilde{x}(t_{k+N_s|k}) = x_{t_{k+N_s|k}}^*, \quad (16b)$$

$$(14c), (14d), \quad (16c)$$

$$u(t) \in \mathcal{U}(p(t)), x(t) \in \mathcal{X}, \quad (16d)$$

$$t \in [t_{k+N_s}, t_{k+N}]. \quad (16e)$$

The formulations (15) and (16) indicate that the short-horizon EMPC $\mathbb{P}_{N_s}$, when equipped with the cost-to-go function as its terminal cost, can replicate the performance of the long-horizon EMPC $\mathbb{P}_N$. However, this cost-to-go function is implicitly defined through the optimisation problem in (16) and is generally not available in closed form. To address this, a parameterized approximation of the cost-to-go function can be adopted, allowing for some acceptable performance loss. The next section introduces a learning-based approach to approximate this terminal cost using feedforward neural networks.

Remark 3.1: A key challenge in solving long-horizon EMPC optimization problems lies in the fact that the battery pack's internal resistance dynamics depends on its future charging and discharging states, which are not available prior to optimization. To address this issue, the proposed approach approximates the battery's future operating states by leveraging the open-loop optimal solution from the previous EMPC optimization. This approximation enables the prediction of resistance dynamics at each time step within the current prediction horizon.

B. The short-horizon EMPC with learning-based terminal cost

To alleviate the online computational burden associated with the original long-horizon problem formulation $\mathbb{P}_N$, an explicit terminal cost $\hat{\mathcal{V}}_f(x, \hat{p})$ can be trained using neural networks to approximate the implicit cost-to-go function $\mathcal{V}_f$ defined as (16). This approximation facilitates the reformulation of $\mathbb{P}_N$ as a short-horizon EMPC problem with N_s , significantly reducing real-time computational demands. We pick the following expression

$$\hat{\mathcal{V}}_f(x, \hat{p}) = \epsilon^2(x, \hat{p}) \times (x - SOC_r)^2, \quad (17)$$

which always remains positive. The inputs for this explicit terminal cost are the state variable x and future power demand $\hat{p} = [p_{t_{k+N_s}}, \dots, p_{t_{k+N}}]$ which is a vector formed by the predicted future vehicle speeds with $N_s \leq \hat{N} \leq N-1$. In order to learn the function $\hat{\mathcal{V}}_f(x, \hat{p})$ from data, we select $p^i = [p_{t_k}, \dots, p_{t_{k+N-1}}]$ and $x_0^i = x_{t_k} \in \mathcal{X}$, for $i \in \mathbb{I}_1^M$. Then, we solve the long-horizon optimization problem $\mathbb{P}_N$ and store the optimal control input $u_0^i = u_{t_k|k}^*$, the corresponding optimal state $x_{N_s}^i = x_{t_{k+N_s|k}}^*$, the optimal initial stage cost $l_0^i = l_e(x_0^i, u_0^i, p_{t_k})$, and the cost-to-go $\mathcal{V}_{N_s}^i = \mathcal{V}_f(x_{N_s}^i, p(t_{k+N_s:k+N-1}))$ calculated by (16), in dataset $\mathbb{D}$.

The $\epsilon(x, \hat{p})$ can be parameterized using a finite-dimensional vector θ based on the collected dataset $\mathbb{D}$, denoted as $\epsilon_\theta(x, \hat{p})$. A neural network is adopted to model this function due to its universal approximation capability, and it is structured as follows:

$$\mathcal{L}(x, \hat{p}; \theta, H) = [A_{H+1} \circ G_{\theta_{H+1}} \circ A_H \circ G_{\theta_H} \circ \dots \circ A_1 \circ G_{\theta_1}](x, \hat{p}) \quad (18)$$

where $\mathcal{L}(x, \hat{p}; \theta, H)$ represents the output of the neural network. Here, H denotes the total number of hidden layers, and h indexes the h th hidden layer, with $h = 1, \dots, H$. The network is constructed as a sequence of layers, where each

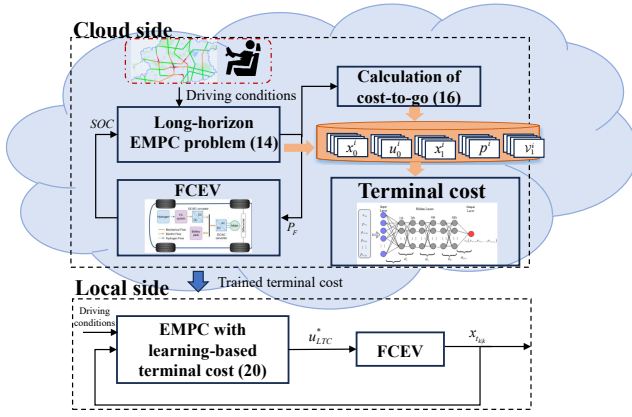


Fig. 4. A schematic illustration of the proposed strategy.

layer applies an affine transformation $\mathcal{G}_{\theta_h}(z_{h-1})$ followed by a nonlinear activation function $\mathcal{A}_h$. Consequently, the output of the h th hidden layer can be expressed as:

$$G_{\theta_h}(z_{h-1}) = w_h z_{h-1} + b_h \quad (19a)$$

$$z_h = \mathcal{A}_h(G_{\theta_h}(z_{h-1})), \quad h \in \mathbb{I}_1^{H+1} \quad (19b)$$

with $z_0 = [x, \hat{p}]$ and $\theta = [\theta_1, \theta_2, \dots, \theta_{H+1}]$ with $\theta_h = [w_h, b_h]$ containing weights $w_h \in \mathbb{R}^{n_{h-1} \times n_h}$ and biases $b_h \in \mathbb{R}^{n_h}$, where n_h stands for a number of neurons in the layer h .

The neural network parameters θ are determined by solving the following finite-dimensional optimization problem:

$$\min_{\theta} \gamma \|\theta\|_2^2 + \frac{1}{M} \sum_{i=1}^M \varphi(\mathcal{V}_{N_s}^i, \hat{\mathcal{V}}_{f\theta}(x_{N_s}^i, \hat{p}^i)), \quad (20a)$$

$$\theta^* := \arg \min_{\theta} \gamma \|\theta\|_2^2 + \frac{1}{M} \sum_{i=1}^M \varphi(\mathcal{V}_{N_s}^i, \hat{\mathcal{V}}_{f\theta}(x_{N_s}^i, \hat{p}^i)), \quad (20b)$$

where φ is a suitably defined loss function.

Once the terminal cost $\hat{\mathcal{V}}_{f\theta^*}(x, \hat{p})$ has been learned, the short-horizon EMPC with learning-based terminal cost, executed on the local side, can be formulated as follows:

$$\min_{u(t) \in S(\Delta)} \sum_{i=0}^{N_s-1} \int_{t_k+i}^{t_k+i+1} l_e(\tilde{x}(t), u(t)) dt + \hat{\mathcal{V}}_{f\theta^*}(\tilde{x}(t_{k+N_s}), \hat{p}) \quad (21a)$$

$$\text{s.t. } \tilde{x}(t_k) = SOC(t_k), \quad (21b)$$

$$(15c), (15d), (15e) \quad (21c)$$

which we denote this optimization problem as $\mathbb{P}_{EMPC-LTC}$. Therefore, the action applied to the FCEV on the local side is denoted as $u_{LTC}^*(t|t_k), \forall t \in [t_k, t_{k+1})$, which is the current input of the solution of optimization problem $\mathbb{P}_{EMPC-LTC}$. Figure 4 shows the overall control structure of the proposed strategy. Besides, the algorithms for dataset establishment and the control procedure on local side at time t_k are summarized in Algorithm 1 and Algorithm 2.

Algorithm 1 Dataset establishment.

Require: $f, g, l_e, \mathcal{U}(p(t)), \mathcal{X}$

Ensure: Dataset $\mathbb{D}$

- 1: Initialization: $\mathbb{D} = \emptyset, i = 1$
- 2: **while** $i \leq M$ **do**
- 3: pick a random initial state $x_0^i = x_{t_k} \in \mathcal{X}$ and its power demand $p^i = [p_{t_k}, \dots, p_{t_k+N-1}]$
- 4: solve the long-horizon EMPC optimization problem $\mathbb{P}_N$ to obtain $U_{t_k}^*$ and $X_{t_k}^*$
- 5: calculate the value of cost-to-go $\mathcal{V}_f^i$ via the optimization problem (16)
- 6: **if** $x_0^i \notin \mathbb{D}$ **then**
- 7: $\mathbb{D} = \mathbb{D} \cup x_{N_s}^i, p^i, \mathcal{V}_f^i, i=i+1$
- 8: **else**
- 9: go back to 3
- 10: **end if**
- 11: **end while**

Algorithm 2 EMPC-LTC Scheme.

Require: $[p_{t_k}, \dots, p_{t_k+N}], \theta^*, \mathcal{L}(x, \hat{p}; \theta^*, H)$

Ensure: u_{t_k}

- 1: Evaluate $\hat{\mathcal{V}}_{f\theta^*}(x, \hat{p})$ via $\hat{p} = [p_{t_k}, \dots, p_{t_k+N}]$ and $\mathcal{L}(x, \hat{p}; \theta^*, H)$
- 2: Solve problem $\mathbb{P}_{EMPC-LTC}$ and obtain $u_{t_k|k}^*$
- 3: Apply $u_{t_k} = u_{t_k|k}^*$ to system (1)
- 4: Set $k = k + 1$ and go back to 1

Remark 3.2: In this proposed method, data are collected for each specific road condition, and a corresponding neural network model is learned to serve as the terminal cost function for short-horizon EMPC. In future work, a large-model-based approach will be employed to construct an EMPC control strategy that encompasses all driving conditions, enabling effective transfer across different scenarios.

C. Probabilistic Validation

Although the neural network can approximate any nonlinear mapping with arbitrary precision, provided a sufficient number of hidden units, the effectiveness of short-horizon EMPC with a learning-based terminal cost should be verified before execution on the local side. According to the universal approximation theorem, the following inequality holds,

$$|u_N^*(t) - u_{LTC}^*(t)| \leq c_e, t \in [t_k, t_{k+1}) \quad (22)$$

for the same state at time t_k , where c_e serves as the upper bound on the approximation error of the control action between the long-horizon EMPC and short-horizon EMPC. To assess the effectiveness of the short-horizon EMPC, i.e., whether it satisfies inequality (22) with a predefined minimum probability, a probabilistic validation procedure is conducted. By generating a sufficient number of uncertain scenarios, the probability that the short-horizon EMPC meets the inequality condition is estimated. The learning-based terminal cost is iteratively trained until this probability exceeds the specified threshold ρ_s .

The following performance indicator is defined to track whether the short-horizon EMPC satisfies inequality (22) along a trajectory for each scenario:

$$\varphi_i := \begin{cases} 1, & \text{if } \|u_N^*(t) - u_{LTC}^*(t)\| \leq c_e, \\ 0, & \text{otherwise} \end{cases}, \quad i = 1, \dots, M_p \quad (23)$$

where M_p denotes the number of independently and identically distributed (i.i.d.) uncertainty scenarios generated for the purpose of probabilistic validation. The probability that $\varphi_i = 1$ is denoted as $\rho := \Pr[\varphi_i = 1]$, and the relationship between this true probability and its empirical estimate $\tilde{\rho} := \frac{1}{M_p} \sum_{j=1}^{M_p} \varphi_j$ can be revealed through the following Hoeffding's inequality,

$$\Pr[|\tilde{\rho} - \rho| \leq \kappa] \geq 1 - \gamma(\kappa, M_p) \quad (24)$$

where $\Pr[|\tilde{\rho} - \rho| \leq \kappa]$ is the probability for $|\tilde{\rho} - \rho| \leq \kappa$, and $\gamma(\kappa, M_p) = 2e^{-2M_p\kappa^2}$. Inequality (24) indicates that the probability of $|\tilde{\rho} - \rho| \leq \kappa$ is at least $1 - \gamma(\kappa, M_p)$, establishing a probabilistic relationship between the empirical risk $\tilde{\rho}$ and the true probability ρ . The short-horizon EMPC is considered effective if it satisfies inequality (22) with a predefined minimum probability ρ_s , i.e., $\rho \geq \rho_s$. To verify this, M_p is gradually increased, and ρ is computed using Hoeffding's inequality with the specified $\gamma(\kappa, M_p)$. The learning-based terminal cost is trained iteratively until the short-horizon EMPC satisfies inequality (22) with at least the desired probability ρ_s .

IV. EVALUATION OF THE PROPOSED EMPC STRATEGY

In this section, the effectiveness of the proposed EMPC with a learning-based terminal cost is validated and analysed using an intelligent networked vehicle test bench. Additionally, real-world driving data collected under actual road conditions further substantiates the hydrogen efficiency and SOC stability achieved by the proposed control scheme. The dynamic optimization algorithm for continuous EMPC, both long horizon and short horizon, is the collocation method, which uses the linear combination of polynomials to approximate the solution of the dynamic system. The EMPC optimization problem is subsequently solved using a gradient-based numerical solver, specifically IPOPT implemented via CasADi in Python 3.7.6. The neural network for terminal cost is developed in PyTorch and trained using the Adam optimizer to solve the learning problem (20).

A. Experimental results of the proposed EMPC

To closely replicate real-world driving conditions, a hardware-in-the-loop (HIL) experiment is conducted on an advanced intelligent connected vehicle test bench, validating the effectiveness of the proposed EMPC strategy. The HIL platform integrates a real driving environment with simulated vehicle dynamics to assess the performance of an actual FCEV. The structure of the HIL experiment setup is illustrated in Figure 5. A virtual traffic environment is established using the SCANer simulation platform, which generates a realistic urban traffic flow scenario. Within this environment, the target vehicle is controlled via a human-machine interface, and

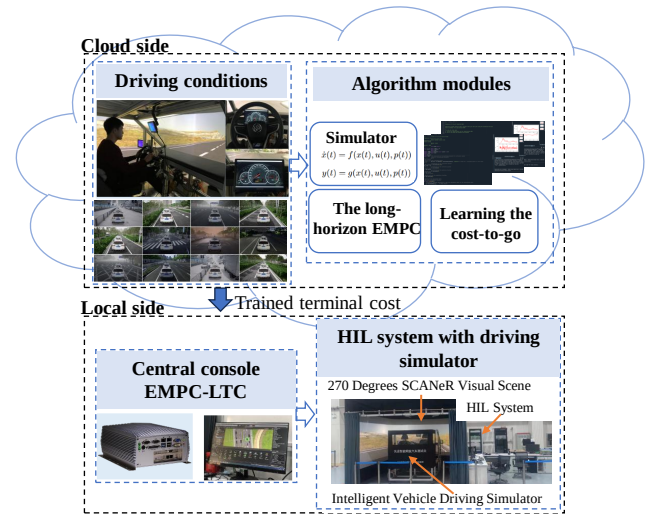


Fig. 5. The HIL experiment platform.

the resulting power demand is transmitted to the powertrain system platform, enabling a closed-loop interaction between the virtual and physical systems for control and optimization verification.

Corresponding to the experiment-based driving conditions, the speed profile and power demand, computed according to (2), are presented in Figure 6. The long-horizon EMPC (14) with prediction horizon $N = 300s$ is executed for collecting the dataset based on Algorithm 1. The prediction horizon for short-horizon EMPC is chosen as $N_s = 50$, and the $\hat{N}$ used for neural network construction is $\hat{N} = 120$. The sampling time is set to be $\Delta = 1s$, and the weighted coefficients for the economic cost function l_e of the EMS are set to be $\gamma_1 = 1$ and $\gamma_2 = \gamma_3 = 10$.

For EMPC-LTC, the forward neural network is incorporated into the EMPC framework to shape the terminal cost which is regarded as a good approximation of the cost-to-go performance of the long-horizon EMPC. The M data samples employed to train the approximate terminal cost function $\hat{V}_{f_{\theta^*}}(x, \hat{p})$ are generated. $M_{train} = 0.8M$ are used as training samples for learning the parameters. The remaining $M_{test} = 0.2M$ samples are reserved to evaluate the performance of the trained neural network model using the normalized root mean squared error (NRMSE). The following function serves as an activation function of the learning-based terminal cost [20],

$$A(x) = \log(1 + x^T x). \quad (25)$$

Besides, a conventional fully connected deep neural network architecture is employed, consisting of three layers with 8, 4, and 1 nodes, respectively. To ensure that the network can adequately approximate the value function while maintaining efficient training, the number of nodes was selected empirically through trial and error, striking a balance between approximation accuracy and computational efficiency. In order to verify the effectiveness of the EMPC with learning-based terminal cost, probabilistic validation is employed with $\gamma = 0.01$ and $\rho_s = 0.97$.

Especially, normalization is undoubtedly an essential consideration in neural network training, as the neural network inputs, i.e., SOC and power demand, differ significantly in their orders of magnitude. It should be noted that normalization functions must be incorporated into the learning-based terminal cost construction to ensure the correctness and reliability of the optimization results.

To evaluate the performance of the proposed EMPC-LTC, a comparison is made with a long-horizon EMPC, which serves as the benchmark for economic assessment. For ensuring a fair comparison, the long-horizon EMPC and the proposed EMPC-LTC are solved using identical computational resources both running on the local side. The optimization results of both strategies are depicted in Figure 7-8. Both controllers demonstrate satisfactory closed-loop economic performance: the fuel cell power remains concentrated around 5kW with only minor fluctuations, and the battery SOC oscillates around the target value of 0.6. Nevertheless, the total computational burden of solving the EMPC-LTC problem is significantly reduced due to its use of a short prediction horizon. The proposed method achieves an reduction in computation time compared to the conventional long-horizon EMPC, highlighting its suitability for real-time implementation.

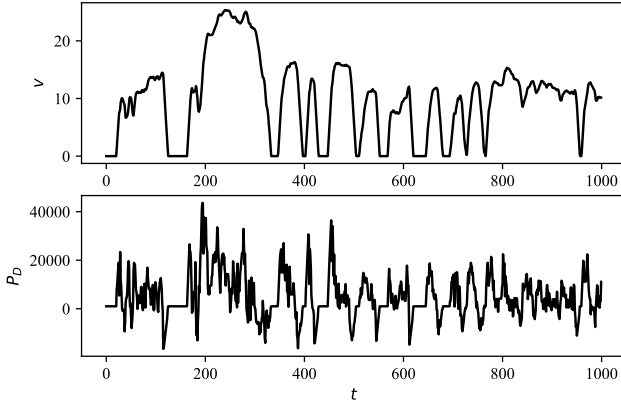


Fig. 6. The speed profile and power demand.

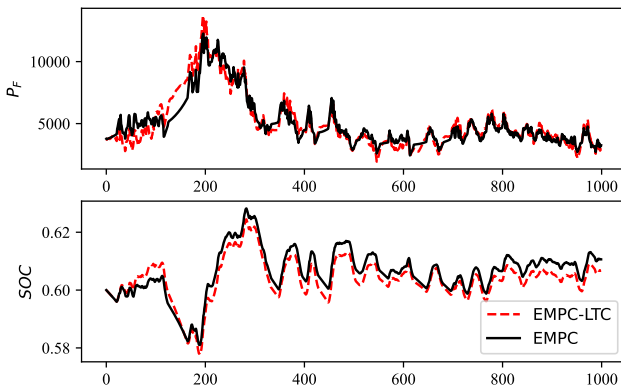


Fig. 7. The optimization results for EMPC and EMPC-LTC.

The effectiveness and robustness of the proposed EMPC-LTC are further validated under real overpass driving condi-

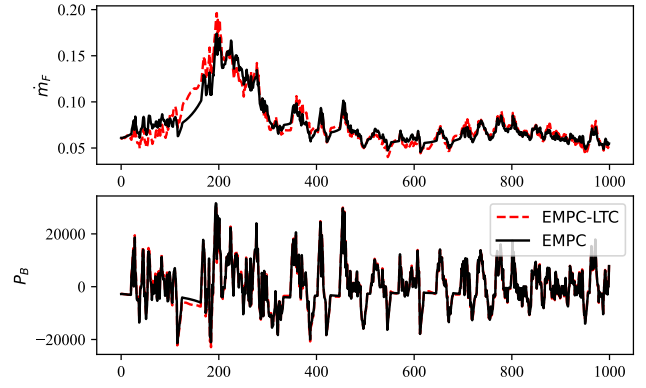


Fig. 8. The optimization results for EMPC and EMPC-LTC.

tions in Wuhan, China, as illustrated in Figure 9-11. The total route spans approximately 32 km, during which the speed trajectory corresponding to preferred driving behaviours is recorded for analysis. The conclusion can be obtained that EMP-LTC has nearly the same economic performance as the long-horizon EMPC, with a lower computational burden.

B. Performance analysis

To quantitatively compare the economic performance and computational efficiency of the two controllers both under experiment-based driving conditions and real overpass driving conditions, define the average closed-loop economic performance, average hydrogen consumption and average SOC tracking error as

$$T_{col} = \frac{1}{N_{sim}} \sum_{i=1}^{N_{sim}} l_e(x(i), u(i)), \quad (26)$$

$$T_{m_F} = \frac{1}{N_{sim}} \sum_{i=1}^{N_{sim}} \dot{m}_F(i), \quad (27)$$

$$T_{soc} = \frac{1}{N_{sim}} \sum_{i=1}^{N_{sim}} (SOC(i) - SOC_r)^2, \quad (28)$$

where N_{sim} is the simulation step, set as 1000. The comparison results are listed in Table II, which presents the experimental results under various driving conditions and weighted coefficient configurations for the economic cost function. Each cell in the table contains two values: the upper value corresponds to the performance of the long-horizon EMPC, while the lower value represents the performance of the proposed short-horizon EMPC-LTC. From the quantitative numerical results and figures, for both experiment-based driving conditions and real overpass driving conditions, we can concluded that:

- 1) The EMPC-LTC achieves closed-loop economic performance comparable to that of the long-horizon EMPC while significantly reducing the computational burden on the local side. This is made possible by offloading part of the performance characterization to the cloud, where a data-driven terminal cost function is learned in advance. This learned terminal cost effectively shortens

TABLE II
EXPERIMENTAL RESULTS WITH DIFFERENT CONFIGURATIONS

Driving conditions	Weighted coefficients	T_{col}	$T_{\dot{m}_F}$	T_{soc}	Computing burdens (sec)
Experiment-based driving conditions	$\gamma_1 = 1, \gamma_2 = 10$	0.0771	0.0761	1.034×10^{-3}	84
		0.0871	0.0873	0.724×10^{-3}	11
	$\gamma_1 = 1, \gamma_2 = 1000$	0.0857	0.0827	3×10^{-3}	86
		0.0866	0.0828	3.026×10^{-3}	15
Real overpass driving conditions	$\gamma_1 = 1, \gamma_2 = 10$	0.136	0.132	3.663×10^{-3}	97
		0.137	0.140	0.91×10^{-3}	21
	$\gamma_1 = 1, \gamma_2 = 1000$	0.141	0.14	1.53×10^{-3}	99
		0.142	0.141	1.43×10^{-3}	19

the required prediction horizon without compromising the optimization performance.

- 2) The EMPC-LTC essentially introduces a novel weighted correction to the squared deviation of the terminal SOC, where the weighting coefficient is determined by the output of a neural network. As a result, in the closed-loop performance of EMPC-LTC, the SOC exhibits smaller fluctuations compared to the long-horizon EMPC. However, this comes at the cost of increased deviation in another performance metric, i.e., hydrogen consumption $\dot{m}_F$.
- 3) When the weighted coefficient for SOC in the economic cost function l_e increases, the closed-loop performance gap between the two controllers narrows. In this case, the advantage of EMPC-LTC becomes more pronounced, as it achieves nearly identical closed-loop economic performance while substantially reducing the online computational burden.

Quantitatively, Table II demonstrates that the proposed EMPC-LTC achieves a computational time reduction of approximately 80–85% compared to the long-horizon EMPC across all test scenarios. In terms of performance, the average total cost T_{clo} remains closely aligned between the two methods, with deviations within 1–2%, indicating that the EMPC-LTC maintains economic optimality despite the shorter horizon. Meanwhile, the SOC deviation T_{soc} is consistently lower for EMPC-LTC, especially under larger SOC weight settings, where it improves by up to 50% compared to the long-horizon baseline. However, this comes with a slight increase in fuel consumption cost.

V. CONCLUSION

This study has proposed a cloud-assisted economic model predictive control with a learning-based terminal cost that explicitly exploits connected-vehicle information to optimize the energy management of FCEVs. The framework delegates a long-horizon EMPC to the cloud to generate data and train a neural-network approximation of the cost-to-go, which is then embedded as a terminal cost in a short-horizon on-board EMPC. A probabilistic validation procedure guarantees that the approximated controller remains within a prescribed performance envelope relative to its long-horizon counterpart.

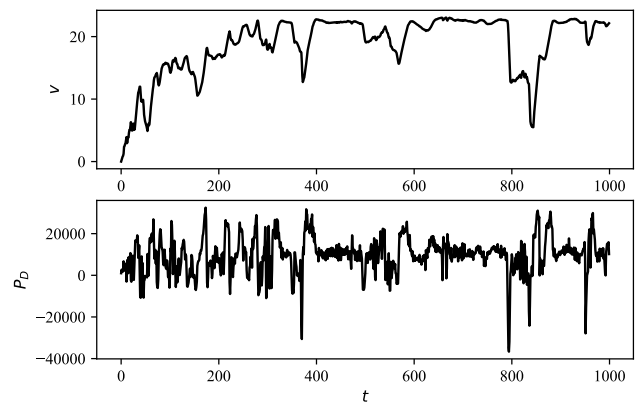


Fig. 9. The speed profile and power demand.

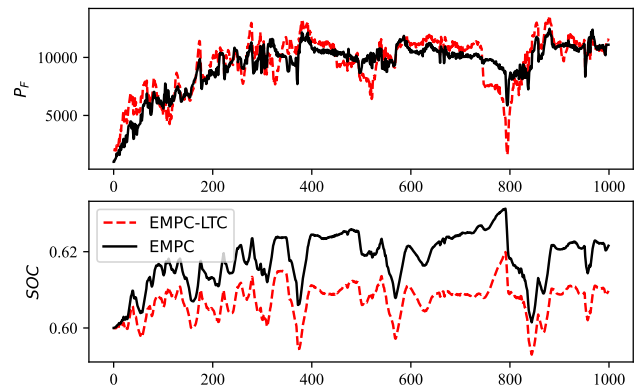


Fig. 10. The optimization results for EMPC and EMPC-LTC.

Hardware-in-the-loop experiments and real-world overpass tests confirm the benefits of the architecture. With the prediction horizon reduced from 300s to 50s, EMPC-LTC maintains hydrogen-consumption and SOC-regulation performance that is nearly the same with the benchmark long-horizon EMPC, yet lowers on-line computational time. These findings demonstrate that EMPC-LTC achieves a favourable balance between closed-loop optimality and real-time feasibility, thereby providing a practical path for deploying advanced economic MPC in connected FCEVs. Future work will build

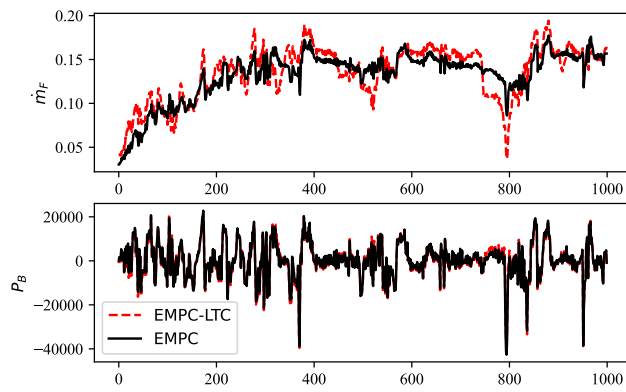


Fig. 11. The optimization results for EMPC and EMPC-LTC.

a universal large-model terminal-cost learner that generalizes across diverse traffic and road scenarios.

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